
Sensor
Raw data
Encoder input
Computation
Properties
(a) Frame camera + RGB encoder
RGB camera
sync, 30–60 fps
frames I_t every 33 ms
H×W×3, motion blur,
static background repeated
same frames
(dense tensor)
CNN / ViT
dense MAC on every pixel,
every frame
temporal res.: frame rate
latency ≥ frame interval
ops: MAC, dense, GPU
(b) DVS + RGB (ANN) encoder
event camera
async per pixel, µs
event stream $e_i = (x_i, y_i, t_i, p_i)$
x–y–t point cloud
red/blue polarity
binned to frame / voxel
grid B×H×W over ΔT
sparsity & timing lost
CNN / ViT (as above)
dense MAC incl. zeros
temporal res.: bin ΔT
latency ≥ window ΔT
ops: MAC, dense, GPU
(c) DVS + SNN
event camera
async per pixel, µs
same event stream
as spike trains per pixel
(raster)
spikes fed directly
(time steps T)
LIF neurons: V integrates,
spike at $V \geq \theta$, reset
AC ops only on spikes
temporal res.: µs–ms
event-driven, low latency
ops: AC, sparse, neuromorphic